Validation_Report

TestID	Contents	Limit	Measured	Margin	Result	Date	Tester
PTO-VRP-000	**Safety validation report — PTO-ECU 200 (build P2)** Results per ISO 25119-4:2018, 6.5. Configuration under test: HW rev C, SW 1.4.0, parameter set V7, T6100 test machine #TM-041 and HIL rig HR-2. The Margin column is a live computed attribute (Limit − Measured) — numeric rows only; qualitative tests carry 0/0. Two first-run failures were analysed and closed: PTO-VAL-009 (fender-jog stop 234 ms > 200 ms — valve pilot orifice tolerance; fixed by parameter V7 jog pre-fill) and PTO-VAL-014 (spurious DTC on 100 µs micro-cuts — input filter time constant corrected in HW rev C). Both re-ran green; regression scope: VAL-004…-007.				Passed	2026-07-24	—
PTO-VAL-001	24/24 invalid patterns rejected; acceptance latency	50	32	18	Passed	2026-07-06	T. Kowal
PTO-VAL-002	Either-channel cut-off verified (halt L1 / halt L2)	0	0	0	Passed	2026-07-06	T. Kowal
PTO-VAL-003	Stuck-conducting detection, demands to detect	1	1	0	Passed	2026-07-07	T. Kowal
PTO-VAL-004	Current-window violation → stop trigger time [ms]	30	22	8	Passed	2026-07-07	T. Kowal
PTO-VAL-005	12/12 power-up scenarios: outputs stayed disabled	0	0	0	Passed	2026-07-08	T. Kowal
PTO-VAL-006	48 resets across states: always OFF at recovery	0	0	0	Passed	2026-07-08	T. Kowal
PTO-VAL-007	Stop chain worst case, −25 °C cold oil [ms]	400	316	84	Passed	2026-07-12	T. Kowal
PTO-VAL-008	Presence loss → stop [ms] (limit 1 s + chain)	1400	1180	220	Passed	2026-07-12	T. Kowal
PTO-VAL-009	Fender-jog stop on release [ms] — re-run with V7	200	158	42	Passed (re-run)	2026-07-15	T. Kowal
PTO-VAL-010	Mode mismatch inhibit / DM-2 entry [ms]	500	410	90	Passed	2026-07-13	T. Kowal
PTO-VAL-011	Overspeed → ramp-down complete [ms]	1000	820	180	Passed	2026-07-13	T. Kowal
PTO-VAL-012	Torque peak ratio vs steady state [—]	1.4	1.27	0.13	Passed	2026-07-16	T. Kowal
PTO-VAL-013	Q&A / WD faults → SAFE [cycles]	2	2	0	Passed	2026-07-14	T. Kowal
PTO-VAL-014	Supply-fault battery: no spurious engagement — re-run rev C	0	0	0	Passed (re-run)	2026-07-18	T. Kowal
PTO-VAL-015	Environmental campaign: functional deviations	0	0	0	Passed	2026-07-21	T. Kowal
PTO-VAL-016	WCET L1 20 ms frame [% budget]	100	62	38	Passed	2026-07-21	C. Moreau
PTO-VAL-017	Corrupt/out-of-range parameter sets rejected	0	0	0	Passed	2026-07-22	T. Kowal
PTO-VAL-018	Field: 60 h, 412 engage/stop events, 0 safety anomalies	0	0	0	Passed	2026-07-24	C. Reid
—	**Conclusion** All 18 validation items passed (2 after corrective action + re-run). Every safety goal is validated end-to-end; measured stop-chain and jog margins are comfortable. Validation documentation per 6.5 complete → input to the assessment and the product release.				Passed	2026-07-24	T. Kowal